databricks

Advance Lakeflow Jobs Features

LECTURE

Handling Task Failures

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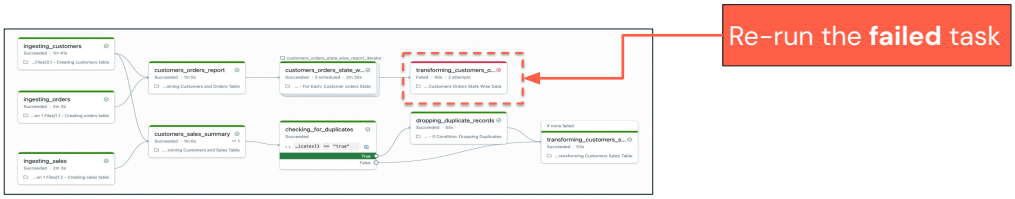
Failure handling isn't just about restarting tasks - it's about building resilient systems that can recover efficiently and maintain data consistency even when components fail.

Handling Task Failures

Repair and Rerun



- **Repair** feature allows to **re-run** and override task parameters
- **Reduces the time** and resources required to recover from unsuccessful job runs
- In case of **Task Failure,** You can:
  - Modify the **task** and run again
  - Modify the **parameters** and run again

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The Repair feature represents a sophisticated approach to failure recovery:

Targeted Recovery: Instead of restarting entire workflows, you can modify specific failed tasks and rerun only what's necessary. This saves significant time and computational resources.

Parameter Override Capability: The ability to modify parameters during repair runs enables you to fix configuration issues, adjust resource allocation, or change processing logic without rebuilding the entire job.

Recovery Scenarios:

- Configuration Fixes: Correct parameter values that caused task failures
- Resource Adjustments: Increase memory or compute resources for tasks that failed due to resource constraints
- Code Updates: Deploy fixes for logic errors and rerun only affected tasks
- Data Quality Issues: Adjust processing logic to handle data quality problems discovered during execution

Cost Efficiency: By rerunning only failed tasks, you minimize unnecessary computation and reduce costs, especially important for large, complex workflows.

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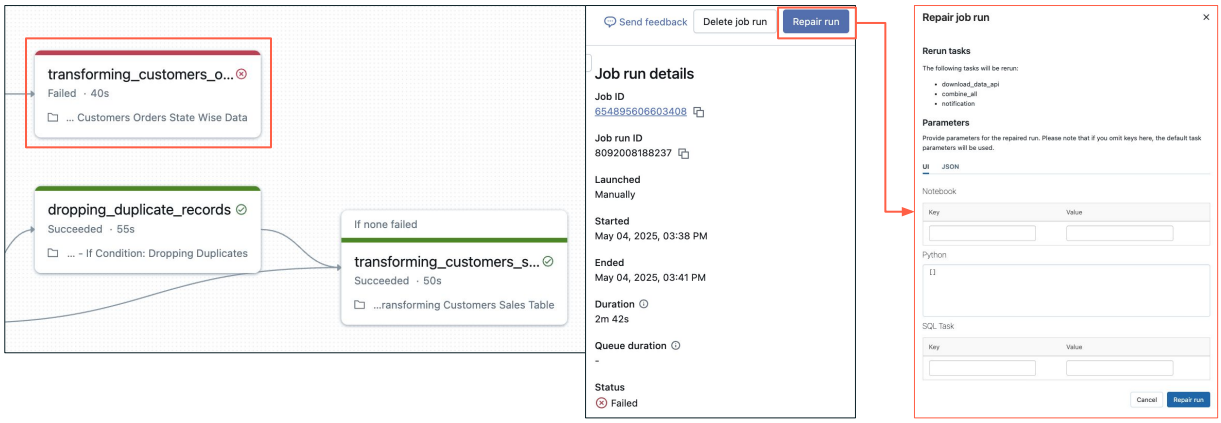

Allowing you to **run only failed tasks**, saving you time and money by only re-running the necessary tasks instead of the entire job

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The selective re-execution capability provides massive operational benefits:

- Resource Optimization: Running only failed tasks instead of entire workflows can reduce recovery time by 80-90% in complex pipelines, saving both time and money.
- Reduced Risk: Smaller recovery operations have less impact on system resources and reduce the risk of cascading failures during recovery attempts.
- Faster Resolution: Teams can respond to failures more quickly when they don't need to wait for entire workflows to complete, improving SLA adherence and business responsiveness.

Please note that fixing a task does not mean fixing a job.

For example: If you pass the wrong parameter and fix it using the repair run feature, you still need to edit that in your job.

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After re-running task, your final job will look like this.

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After repair operations, the job run history clearly shows the recovery process:

Audit Trail: Complete visibility into what was repaired, when, and by whom, providing essential information for troubleshooting and process improvement.

Success Validation: Clear indication of which tasks were recovered successfully, enabling confidence in the repair process.

Learning Opportunities: Historical repair data helps teams identify patterns in failures and improve initial job design to prevent future issues.

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Thank you for completing this lesson and continuing your journey to develop your skills with us.